SAR-Optical Feature Fusion for Crop-Type Classification

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***Abstract*—Crop-type mapping from satellite imagery supports agricultural monitoring, but optical data is often unavailable during the growing season because of cloud cover, while Synthetic Aperture Radar (SAR) data is weather-independent but noisier and harder to interpret. This project investigates whether feature-level fusion of classical, hand-crafted optical and SAR features improves crop-type classification compared with either modality alone. We use the YieldSAT dataset (Sentinel-2 time series with crop labels) for Germany and Uruguay, and acquire matching Sentinel-1 VV and VH data through Google Earth Engine, Optical features include vegetation indices (NDVI, NDRE, NDMI, EVI) and GLCM texture; SAR features include backscatter statistics, the VV/VH ratio and texture. So far, we have analysed the dataset structure, computed NDVI time series for rapeseed, wheat and soybean from sample pixels, observed missing optical acquisitions, and retrieved a Sentinel-1 time series for a test location. Because Uruguay contains only soybean, the classification task is rapeseed versus wheat in Germany. We have selected Random Forest and a Support Vector Machine (SVM) as classifiers, which will be trained on optical-only, SAR-only and fused features and evaluated by accuracy, F1-score and confusion matrices.**

***Index Terms*—SAR, Sentinel-1, Sentinel-2, crop classification, feature fusion, GLCM, Random Forest**

I. Introduction

Accurate crop-type maps are essential for agricultural monitoring, yield estimation and food-security planning. Optical satellite sensors such as Sentinel-2 capture spectral reflectance that is directly related to vegetation health, and are therefore the standard input for crop mapping. However, optical imagery is often unavailable during critical stages of the sample data we examined, several acquisition dates already contain missing optical values, Synthetic Aperture Radar (SAR) sensors such as Sentinel-1 are not affected By clouds, but SAR backscatter is noisy because of speckle and is less directly interpretable in terms of crop biophysical properties.

Combining the two modalities is therefore attractive. Kpienbaareh et al. [4] classified crop types and land cover in northern Malawi using a Random Forest on Sentinel-1, Sentinel-2 and PlanetScope data. Sentinel-1 alone reached an overall accuracy below 50%, Sentinel-2 alone about 72–74%, and the fused stack above 85%, an improvement of 11.7 to 13.5 percentage points over Sentinel-2 alone [4]. That study fused data by stacking image layers; in this work we instead study fusion at the feature level, using classical hand-crafted features rather than deep learning.

We use YieldSAT [**?**], a multimodal benchmark providing Sentinel-2 time series, crop labels and yield maps. We work with its Germany and Uruguay subsets, which contain rapeseed and wheat (Germany) and soybean (Uruguay), and we acquire matching Sentinel-1 data through Google Earth Engine, since YieldSAT does not include SAR imagery. As Uruguay contains a single crop, the classification task is rapeseed versus wheat in Germany, and the Uruguay data is used for time-series comparison. We select Random Forest, the classifier used in [4], and a Support Vector Machine as two complementary classical models, and compare them across optical-only, SAR-only and fused settings. Our objective is to determine whether fusing optical and SAR features improves this classification compared with either modality alone.

II. Methodology

A. Dataset

We use the YieldSAT [**?**] preprocessed release: one NetCDF file per country with pixel-level Sentinel-2 time series (24 time steps, 10 m) and crop labels. Germany has 299 fields (111 rapeseed, 188 wheat) and Uruguay 572 soybean fields. Only the Sentinel-2 bands B02, B04, B05, B08 and B11 and the crop label are used. Digital numbers are divided by 10^4 to obtain reflectance (standard Sentinel-2 L2A scaling, assumed here), and missing time steps are excluded. At this stage we analysed small pixel samples.

B. Sentinel-1 Acquisition

As YieldSAT contains no SAR data, Sentinel-1 GRD images (IW mode, VV and VH) will be retrieved through Google Earth Engine for each field and for the season between seeding and harvest, and a speckle filter (e.g., Lee) will be applied. A first test for one location returned 116 images. The SAR-only and fused settings depend on obtaining the field centroids, which are not included in the preprocessed files.

C. Features

Optical features are NDVI, NDRE, NDMI and EVI, where

$$\mathrm{NDVI} = \frac{\rho_{08} - \rho_{04}}{\rho_{08} + \rho_{04}} \tag{1}$$

TABLE I
PEAK MEAN NDVI OF THE SAMPLED PIXELS

Crop (country, year)	Pixels	Peak NDVI	Date of peak
Rapeseed (DE, 2020)	4	0.67	23 Jun 2020
Wheat (DE, 2019)	4	0.83	24 Jul 2019
Soybean (UY, 2019)	5	0.87	20 Mar 2019
Soybean (UY, 2020)	5	0.89	24 Jan 2020

and ρ_k is the reflectance of band Bk. NDRE and NDMI replace B04 with B05 and B11, and EVI additionally uses B02. Each index is summarised per field by its mean, standard deviation and maximum over the season, together with GLCM texture. SAR features are the mean and variance of VV and VH, the VV/VH ratio and GLCM texture.

D. Classification and Evaluation

Optical and SAR features are concatenated per field (feature-level fusion) after standardisation fitted on the training set. Random Forest (also used in [4]) and an RBF-kernel SVM have been selected; deep learning is outside the scope. Both will be trained on optical-only, SAR-only and fused features, using a field-wise train/test split to avoid leakage, class weights for the wheat/rapeseed imbalance, and accuracy, F1-score and confusion matrices as metrics.

III. RESULTS

At this stage, the results are limited to the sample data and a Sentinel-1 retrieval test; classifier results are not yet available.

A. NDVI Time Series

The sampled pixels include rapeseed (Germany, 2020), wheat (Germany, 2019) and soybean (Uruguay, 2019 and 2020). The samples contain 18 pixels from four fields. Soybean NDVI rises to a maximum of about 0.87–0.89 between January and March and drops in April, wheat reaches about 0.83 in June–July, and rapeseed stays lower and flatter, with a maximum mean NDVI of about 0.67 (Table I).

B. Missing Optical Observations

In the German rapeseed sample, 3 of the 12 dated acquisitions had no valid optical values for any of the 50 sampled pixels, and on one date only 21 of the 50 pixels had valid values, possibly because of cloud masking.

C. Sentinel-1 Retrieval Test

A test query for one location in Germany over April–September 2021 returned 116 Sentinel-1 GRD images (IW mode) with VV and VH bands and the incidence angle, confirming that SAR data can be acquired through Google Earth Engine.

IV. CONCLUSION

This report presented the progress on SAR–optical feature fusion for crop classification using YieldSAT. We analysed the dataset structure, derived NDVI time series for rapeseed, wheat and soybean, observed missing optical acquisitions, tested Sentinel-1 retrieval through Google Earth Engine, and selected Random Forest and SVM for rapeseed versus wheat classification in Germany. Next, we will obtain the field centroids, read the full Germany data, retrieve and filter Sentinel-1 images, extract optical and SAR features, and compare optical-only, SAR-only and fused classification using accuracy, F1-score and confusion matrices.

REFERENCES

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